1️⃣ Manipulate numbers, graphical and statistical data (Item 7)

Relevance: ★★★★★ (Most critical)

Description:

> Manipulate numbers, graphical and statistical data, algebraic expressions and equations. employ

How it appears in the problem:

- Performing numerical operations with **distance, time, and speed**
- Using the formula

$$v = \frac{distance}{time}$$

This involves:

- Unit conversion (minutes → hours)
- Ratio calculation
- Numerical comparison

This is the **core mathematical competency** required for this problem.

2️⃣ Perform a simple calculation (Item 1)

Relevance: ★★★★★

Description:

> Perform a simple calculation. employ

Manifestation:

The key steps are:

- Calculate two average speeds
- Compare their values

This represents a **typical basic computation task**.

3️⃣ Understand and utilise constructs based on definitions and rules (Item 5)

Relevance: ★★★★☆

Description:

> Understand and utilise constructs based on definitions, rules and formal systems, including employing familiar algorithms. employ

Manifestation:

Students must understand the **definition of average speed**

$$\text{average speed} = \frac{\text{distance}}{\text{time}}$$

and correctly apply this rule.

4️⃣ Use a multi-step procedure leading to a mathematical solution (Item 12)

Relevance: ★★★★☆

Description:

> Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation. employ

Solution process:

1. Read and interpret the contextual information
2. Calculate the speed for the first interval
3. Calculate the speed for the second interval
4. Compare the results
5. Select the correct option

Thus, the problem requires a **multi-step reasoning process**.

5️⃣ Use an understanding of the context to guide solving (Item 13)

Relevance: ★★★☆☆

Description:

> Use an understanding of the context to guide or expedite the mathematical solving process.
> employ

Manifestation:

The context involves:

- A bicycle
- A speedometer
- A travel trip

Students need to understand the real-world meaning of:

- Distance
- Time
- Average speed

However, the mathematical complexity remains relatively basic.

6️⃣ Make sense of and use representations (Item 10)

Relevance: ★★☆☆☆

Description:

> Make sense of, relate and use a variety of representations when interacting with a problem.
> employ

Manifestation:

Although the problem does not include graphs or tables, students still need to convert:

Verbal description → Mathematical expressions

This represents a basic **representation transformation**.

1️⃣ Manipulate numbers, graphical and statistical data…（第7项）

相关性：★★★★★（最核心）

能力描述：

> Manipulate numbers… algebraic expressions and equations employ

体现：

- 对 **距离、时间、速度**进行数值运算
- 使用公式

$$v = \frac{distance}{time}$$

涉及：

- 单位换算（分钟 → 小时）
- 比例计算
- 数值比较

这是本题**最核心能力。**

2️⃣ Perform a simple calculation（第1项）

相关性：★★★★★

能力描述：

> Perform a simple calculation employ

体现：

题目核心步骤就是：

- 计算两个平均速度
- 再进行大小比较

属于典型的 **基础计算型任务。**

3️⃣ Understand and utilise constructs based on definitions…（第5项）

相关性：★★★★☆

能力描述：

> Understand and utilise constructs based on definitions, rules… including employing familiar algorithms employ

体现：

需要理解 **平均速度定义**

$$\text{average speed} = \frac{\text{distance}}{\text{time}}$$

并按规则应用。

4️⃣ Use a multi-step procedure leading to a mathematical solution（第12项）

相关性：★★★★☆

能力描述：

> Use a multi-step procedure leading to a mathematical solution employ

解题步骤：

1️⃣ 读取情境信息
2️⃣ 计算第一段速度
3️⃣ 计算第二段速度
4️⃣ 比较结果
5️⃣ 选择选项

属于**多步骤推理计算问题。**

5️⃣ Use an understanding of the context to guide solving（第13项）

相关性：★★★☆☆

能力描述：

> Use an understanding of the context to guide solving employ

体现：

题目背景是：

- 自行车
- 速度计
- 行程

学生需要理解：

- 距离、时间、速度的现实意义
- “平均速度”的概念

但数学难度仍较基础。

6️⃣ Make sense of and use representations（第10项）

相关性：★★☆☆☆

能力描述：

> Make sense of, relate and use a variety of representations employ

体现：

虽然题目没有图表，但学生需要：

- 从 **文字情境 → 数学表达式**

属于简单的**表征转换。**